

ECOSYSTEM DISRUPTION PLAYBOOK

Sustained Degradation of the Russia-Linked Ransomware Ecosystem

Developed by Reno

OPERATOR SNAPSHOT

Mission. Sustained ecosystem degradation, not episodic takedowns. Success is measured by compounding friction over time: actors spending more on security and reconstitution, Russian institutions treating criminals as internal liabilities, protection relationships dismantled, ransom volume and operational tempo declining across multi-quarter windows.

The 4 Phases

Phase	Name	Objective	Key Outputs
1	Ecosystem Mapping	Build dependency-linked map across financial, infrastructure, human terrain, protection layer, and state-interaction layers	Dependency graph, choke point inventory, substitutability assessment, protection relationship map
2	Pressure Alignment	Identify which Russian institutions are susceptible to which levers; align pressure to internal contradictions including manufactured contradictions via the protection layer track	Agency alignment matrix, referral targeting plan, reframing strategy, officer liability candidates
3	Cost Imposition	Apply coordinated pressure across financial, infrastructure, legal, social, and protection layer vectors to impose compounding costs	Sanctions actions, infrastructure takedowns, domestic referrals, underground trust disruption, officer exposure operations
4	Sustainment	Monitor adaptation, prevent reconstitution, reapply pressure to emergent nodes and successor protection relationships (continuous, never ends)	Reconstitution tracking, KPI dashboards, updated pressure actions, protection relationship succession map

Critical Do / Don't Rules

DO	DON'T
Lead with domestic harm framing (tax fraud, organized crime, harm to Russian citizens)	Lead with geopolitical harm framing (activates protection reflexes)
Sequence low-backfire-risk actions first; disrupt protection relationship before or simultaneously with the criminal actor	Lead with public attribution or extradition requests (hardens protection, not suppression)
Monitor before takedown; use seizure as a data collection event and feed follow-on actions	Treat a takedown as an endpoint: without sustained follow-on pressure, actors reconstitute

DO	DON'T
	within 30-90 days
Exploit internal Russian institutional contradictions (MVD vs. FSB; FNS financial exposure; FSB factional competition; CBR 115-FZ friction)	Seek coordinated Russian cooperation at the strategic level (not achievable; signals foreign ownership of the case)
Measure every action: define expected effect before acting; log observed results and time-to-reconstitution	Run disruption operations without measurement (narrative without accountability)

SECTION 1 | PURPOSE & SCOPE

What This Document Is

This document proposes a framework for sustained disruption operations targeting the Russia-linked ransomware and cybercrime ecosystem, including the protection infrastructure (Russian state officers and institutional relationships) that insulates top-tier actors from enforcement. It is offered as analytic input for interagency partners with operational or analytic roles.

- A proposed phased framework for ecosystem-level disruption
- A strategic reference for pressure alignment across domestic Russian institutions
- A guide for avoiding engagement patterns that trigger protection reflexes
- A methodology for identifying and dismantling FSB officer protection relationships
- A tool and partner reference for capability targeting

What This Document Is Not

- A policy document or legal authority
- A replacement for agency-specific operational protocols
- A senior leadership orientation document: this is analytic input for operational planning
- A directive for operational units: analytic recommendations require appropriate authority before action
- A static product: it must evolve as the ecosystem adapts

This document draws on structural analysis of Russian agency behavior, observed enforcement outcomes, and analytic assessments of the ransomware ecosystem. Confidence levels are labeled throughout. Realpolitik incentives, not normative expectations, drive the analytical framework.

Scope Limitations

Three categories of actor fall outside this framework's intended scope and should be flagged through separate channels if encountered:

- GRU-linked actors and wartime-absorbed criminal infrastructure. The GRU leverages criminal actors for military and geopolitical operations and maintains different protection dynamics than the FSB. More critically, actors who have been operationally absorbed into GRU or FSB wartime operations occupy a different legal and policy environment. This framework does not address them. Indicators of wartime state absorption (actors receiving operational tasking against NATO or Ukrainian infrastructure, or actors whose targeting patterns align with active Russian military objectives) should be flagged separately.
- Non-Russia CIS actors with independent state protection. This framework's institutional leverage points (FSB, MVD, FNS, CBR) are Russian. Actors operating under Kazakh, Belarusian, or other CIS state protection require jurisdiction-specific analysis. Note: Kazakhstan has functioned as an actual enforcement partner on specific cases and its enforcement dynamics differ meaningfully from Russia's, Kazakhstani actors without Russian state protection may be accessible through local law enforcement channels. Belarus is operationally equivalent to Russia (see Annex A). Other CIS jurisdictions require case-by-case assessment.
- SVR-linked actors. The SVR's relationship with the cybercriminal ecosystem is more limited than the FSB's but is not zero, particularly at the financial fraud and intelligence nexus. This framework does not address SVR-protected actors.

SECTION 2 | CORE THESIS

The Russia-linked ransomware ecosystem is resilient by design. Individual actors, infrastructure nodes, and even ransomware brands reconstitute quickly after episodic disruptions. Takedowns create friction but not degradation unless compounded over time.

The correct strategic model is sustained cost imposition across multiple ecosystem layers simultaneously: financial, infrastructural, human terrain, protection layer, and domestic Russian institutional. When pressure is applied in coordination across these layers, actors face compounding costs that cannot be absorbed through simple reconstitution.

Scope qualification: the compounding friction thesis is structurally sound but rests on an assumption of sustained, coordinated multi-node pressure that has no confirmed historical precedent at this scale. Available evidence — including the 2024 ransomware revenue decline — is consistent with the model but reflects partial implementation across a subset of vectors, not a full coordinated execution. The thesis should be treated as an operational framework and planning model, not an empirically validated outcome. Confidence in the model's logic is high; confidence that full coordinated execution is achievable within current interagency and allied coordination constraints is moderate. [ANALYST INFERENCE]

A critical layer absent from many disruption frameworks is the protection infrastructure itself. FSB officers who recruit, co-opt, and shield cybercriminal actors are not passive bystanders. They are load-bearing nodes in the ecosystem. Disrupting the protection relationship is as operationally important as disrupting the criminal infrastructure it shields.

Episodic Takedowns	vs.	Sustained Disruption
Single actor or infrastructure target		Multiple simultaneous pressure vectors
Rapid reconstitution within 30-90 days		Compounding costs across financial, infra, human, and protection layers
Creates narrative, not lasting friction		Forces tradeoffs between criminal operations and regime stability
No measurable ecosystem effect		Measurable ecosystem health degradation over time
Protection layer untouched		Protection relationships dismantled, not worked around

► CORE MISSION

The objective is not episodic takedowns. It is sustained ecosystem degradation: increasing operational friction, raising domestic risk for criminal actors, forcing alignment of Russia's internal institutional incentives against cybercrime, and dismantling the protection infrastructure that insulates top-tier actors from enforcement. Success is measured by compounding pressure, not individual arrests.

Defining Success

- Increased operational friction: actors spending more time and money on security, reconstitution, and vetting

- Rising domestic risk: Russian institutions treating cybercriminal actors as internal liabilities, not tolerated assets
- Protection layer erosion: specific FSB officers and factions whose protection relationships become institutional liabilities
- Forced tradeoffs: criminal operations creating regime-level costs that cannot be ignored
- Measurable ecosystem contraction: victim counts, ransom volume, and operational tempo declining over multi-quarter windows

The strategic principles that follow govern all phases of this framework. Violating them (particularly the engagement triggers) consistently produces the opposite of the intended effect.

SECTION 3 | STRATEGIC PRINCIPLES

3.1 Lead with Domestic Harm Framing

Russian institutions respond to internal threats, not external ones. Cybercriminals must be reframed as threats to Russian financial stability, domestic public trust, and regime legitimacy, not as foreign adversaries. Framing that emphasizes geopolitical harms activates protection reflexes and is counterproductive. Confidence: CREDIBLE

- Emphasize harm to Russian citizens, businesses, and financial infrastructure
- Use tax irregularities, currency violations, and fraud charges (not cyber-specific charges) where possible
- Surface financial inconsistencies (lifestyle vs. declared income) through FNS referrals
- Avoid framing that allows actors to be cast as resisting foreign interference

3.2 Exploit Internal Contradictions, Including Within FSB

Russian agencies do not operate in unified alignment. The FSB, GRU, MVD, and FNS frequently have competing institutional interests. Effective pressure identifies and exploits these contradictions rather than seeking coordinated Russian cooperation, which is not achievable at the strategic level. Confidence: CREDIBLE

Critically, FSB itself is not a monolith. It contains competing factions, officers with divergent financial interests, and internal competition for political favor. The officer liability track (Section 9) rests on this reality: the goal is not to activate a unified FSB response, but to create conditions where specific FSB factions or leadership figures (who have their own institutional incentives) find it in their interest to act against a specific officer. This is a different and more achievable objective than FSB cooperation.

- MVD Department K has arrest incentives that FSB does not always override for mid-tier actors
- FNS modernization objectives can be leveraged to surface financial anomalies independent of RIS direction
- Rosfinmonitoring operates under Egmont obligations that create exposure even when political will is absent
- CBR 115-FZ creates non-attributable friction without requiring prosecutorial intent
- Within FSB: officers whose protected criminal assets have become attribution risks may face internal pressure from competing officers or factions, this internal friction is exploitable

► MANUFACTURING CONTRADICTIONS

Contradictions are not only exploited. They can be manufactured. The protection layer track (Section 9) operationalizes this: financial exposure operations that surface an FSB officer's criminal protection relationships to competing officers or FSB leadership create contradictions where none previously existed. When an officer's krysha relationship becomes visible to FSB leadership as an attribution risk or domestic embarrassment, the institution's factional dynamics and self-preservation instincts can activate against that officer. You are not waiting for contradictions to exist, you are generating

them through targeted analytic and exposure work.

3.3 Sequence for Minimum Backfire Risk

Actions that trigger protection reflexes (particularly public attribution and extradition requests) harden actor protection rather than undermining it. Pressure sequencing must account for this dynamic. Confidence: CREDIBLE

- Low backfire risk first: financial exposure, infrastructure provider pressure, underground trust node disruption
- Mid-risk second: domestic agency referrals through non-intelligence channels (FNS, MVD)
- Avoid triggering actions unless actor is already isolated from state protection
- Full backfire risk reference: Section 8

3.4 Measure Everything

Disruption operations without measurement become narrative. Every pressure action should have a defined expected effect and a mechanism for observing actual outcomes. Adaptation by actors is itself signal, log it. Confidence: CREDIBLE (as principle); implementation quality varies by resourcing

- Define expected effect before taking action
- Log pressure applied, observed response, and time-to-reconstitution for every major action
- Track ecosystem health KPIs on a recurring basis (see Section 11)
- Use adaptation patterns to predict next-state ecosystem behavior

SECTION 4 | PHASE FRAMEWORK

The four phases are sequential at ecosystem scale but overlapping at the actor level. Phase 4 (Sustainment) begins as soon as the first pressure actions are taken and never ends.

Phase	Name	Primary Objective	Key Outputs
1	Ecosystem Mapping	Build a complete, dependency-linked map of the ecosystem across financial, infrastructure, human terrain, protection layer, and state-interaction layers	Dependency graph, choke point inventory, substitutability assessment, protection relationship map
2	Pressure Alignment	Identify which Russian domestic institutions are susceptible to which levers, and align foreign pressure to exploit internal contradictions, including manufactured contradictions via the protection layer track	Agency alignment matrix, referral targeting plan, reframing strategy, officer liability candidates
3	Cost Imposition	Apply coordinated pressure across financial, infrastructure, legal, social, and protection layer vectors to impose compounding costs	Sanctions actions, infrastructure takedowns, domestic referrals, underground trust disruption, officer exposure operations
4	Sustainment	Monitor adaptation, prevent reconstitution, reapply pressure to emergent nodes	Reconstitution tracking, KPI dashboards, updated pressure actions

4.1 Phase 1: Ecosystem Mapping

Before pressure can be applied effectively, the ecosystem must be understood as an interdependent system, not a collection of individual actors. Mapping focuses on nodes, dependencies, choke points, and substitutability. The protection layer is a mandatory fifth mapping domain.

Financial Layer

- Wallet clusters and attribution: primary wallets, layering wallets, mixing endpoints
- Cash-out infrastructure: OTC brokers, crypto exchanges (VASP compliance gaps), payment aggregators
- Mule networks: recruitment patterns, geographic concentrations, lifecycle
- Front companies and laundering fronts: registration, banking relationships, beneficial ownership
- Anomalous outflows: payments that do not fit criminal operational cost profiles, these are protection payment candidates (see Section 9 and Section 10)

Key question: where do funds become fiat, and who touches them?

Infrastructure Layer

- Hosting and BPH providers: registrar, nameserver, ASN, upstream transit
- Domain infrastructure: registration patterns, DNS clustering, CDN/DDoS protection dependencies
- Payment acceptance: what payment rails accept criminal proceeds at each node
- Upstream dependencies: who hosts the BPH providers? Who sells IP space to the ASNs?

Key question: what is the minimum set of upstream providers whose removal collapses multiple downstream nodes?

Human Terrain

- Technical roles: malware developers, IAB operators, affiliate managers, crypters
- Governance and trust roles: escrow operators, forum administrators, arbitrators, reputation managers
- Financial roles: cash-out brokers, mule recruiters, accountants, front company directors
- State linkages: which actors have RIS recruitment history, krysha relationships, or prior enforcement contact?

Protection Layer

- Map which FSB officers or factions are actively handling, recruiting, or shielding which actors
- Document protection payment flows: anomalous outflows from criminal actor finances that represent krysha fees or bribe payments
- Identify protection relationship tenure: how long has the relationship existed, and has it ever shown signs of strain?
- Map officer personal financial exposure: Western assets, family holdings, real estate, business interests
- Cross-reference officer travel patterns against viable third-country arrest jurisdictions (Annex A)

Key question: for each protected actor, who is the officer or faction, what is the payment relationship, and what would make that officer a liability to FSB leadership or competing factions?

Substitutability Assessment

Substitutability is not a mapping output, it is a targeting input. Build substitutability assessments before disruption actions, not after. For each critical role:

- Identify the likely replacement actor within 30 days of disruption, name them, map them, pre-position pressure on them
- Assess recruitment pipeline depth: is the replacement pool thin (high substitutability cost) or deep (low cost)?
- For protection relationships: who is the backup officer if the primary is removed? Is there one? Absence of a backup officer is a high-value disruption indicator

- Document succession signals in underground forums: who is building reputation that positions them for role succession?
- Pre-position pressure on identified successors so disruption of the primary does not create a clean handoff

► SUBSTITUTABILITY FAILURE PATTERN

The most common operational error: disrupting a node without pre-positioned pressure on the identified successor. The successor steps in cleanly, reconstitution accelerates, and the operation produces a leadership transition rather than ecosystem degradation. Map the successor first. Apply pressure to the successor simultaneously with or before disrupting the primary.

State Interaction Layer

- Map which Russian agencies are currently tolerating, exploiting, or suppressing which actors
- Identify actors without current state protection (highest-value disruption targets)
- Identify contradictions: actors whose behavior creates friction with agencies that could be activated
- Cross-reference against enforcement outcomes and known recruitment histories

4.2 Phase 2: Pressure Alignment

Alignment means identifying which levers work through which Russian institutions, sequencing referrals to exploit internal contradictions, and ensuring domestic framing is in place before any action is taken. The full Russian agency reference is in Section 7.

Priority Channels: Confidence CREDIBLE

- MVD Department K: mid-tier actors without active RIS protection; responsive to domestic harm framing and arrest metrics incentives
- FNS: financial exposure and lifestyle inconsistencies; no arrest power but feeds downstream enforcement
- Rosfinmonitoring: laundering infrastructure mapping; Egmont obligations create exposure pathways independent of political will
- CBR (115-FZ): non-attributable financial friction without prosecutorial threshold; disrupts infrastructure without targeting individuals

Secondary Channels, Confidence: CREDIBLE (conditional)

- FSB factions: reachable when a specific officer or faction has reason to act against a competing officer's protected asset, or when an actor has defied or embarrassed the service. Not reachable through standard channels. Reachable through manufactured liability (Section 9)
- SKR (Investigative Committee): activates only on elite political signal; not accessible through routine referrals

- Prosecutor General: downstream packager, not an initiator; useful for reframing charges once a case is politically elevated

Reframing Strategy

- Tax fraud and undeclared income, not hacking charges
- Organized crime (Article 210), applicable when group structure, hierarchy, and economic benefit are documented
- Harm to Russian citizens or financial stability, not Western victim framing
- Financial irregularities exposing political patrons, creates incentive for krysha relationships to weaken

4.3 Phase 3: Cost Imposition

Cost imposition operates across five simultaneous vectors. Actions within each vector should be sequenced from lowest to highest backfire risk and coordinated across vectors where possible.

Vector 1: Financial Pressure

- Wallet sanctions (OFAC/OFSI/EU): designate wallets with documented criminal attribution
- Exchange KYC pressure: engage VASPs with weak AML compliance; flag suspicious clusters
- Correspondent banking exposure: surface laundering routes that touch Western correspondent banks
- CBR referrals: flag suspicious domestic transaction patterns through Rosfinmonitoring-to-CBR pipeline
- OTC broker exposure: identify and designate key OTC nodes that serve as cash-out bottlenecks

Vector 2: Infrastructure Pressure

- Upstream provider engagement: registrars, hosting companies, DNS providers, CDN/DDoS services
- ISP and ASN notifications: abuse notifications to upstream transit providers for BPH-linked ASNs
- Takedown coordination: infrastructure seizures timed to financial actions to prevent emergency reconstitution
- Roskomnadzor friction: platform restriction requests that disrupt criminal communications channels

Vector 3: Legal and Non-Cyber Charges

- Tax and financial charges: work through FNS exposure to generate domestic enforcement basis
- Currency violations: undeclared foreign assets and illegal currency operations

- Article 210 (criminal organization): applicable when group structure, hierarchy, and economic benefit are documented
- Corruption and bribery: where krysha relationships involve documented payments to officials
- Drug and travel violations: non-cyber charges usable in third-country jurisdictions for arrest during travel

Vector 4: Underground Social Infrastructure

- Trust node disruption: escrow operators, arbitrators, and forum administrators are non-technical but ecosystem-critical
- Reputation system attacks: discrediting actors within underground markets creates internal friction
- Recruitment pipeline disruption: targeting mule recruitment networks raises operational costs
- Communication platform pressure: degrade Telegram channels and forums used for coordination
- Escrow and arbitration record exploitation: disputes between criminal actors and protection relationships surface in arbitration, these records contain protection payment documentation and officer contact methods

Vector 5: Protection Layer Pressure

Full methodology in Section 9. Summary:

- Anomalous outflow analysis: identify protection payment flows from criminal actor finances using the crypto-to-fiat methodology (Section 10)
- Officer financial exposure: map Western asset holdings, family financial relationships, lifestyle inconsistency, feed to FNS and investigative journalism channels
- Manufactured contradiction generation: surface officer-criminal financial relationships to FSB institutional awareness through non-attributable channels, exploiting internal factional competition, not seeking unified FSB cooperation
- Simultaneous officer + criminal network sanctions: batch designations that land on the officer and their protected network simultaneously, eliminating reconstitution window
- Third-country arrest development: pre-position legal frameworks in viable European jurisdictions for officers and former officers with documented travel patterns (Annex A)

4.4 Phase 4: Sustainment

Sustainment is not a final phase, it is a continuous operational posture. Disrupted ecosystems reconstitute unless pressure is reapplied to emergent nodes.

Reconstitution Monitoring

- Track infrastructure reappearance: new domains, ASNs, and hosting providers linked to disrupted actors

- Monitor wallet re-activation: new wallets receiving funds from known criminal clusters
- Track actor rebranding: name changes, new affiliate programs, forum re-registration
- Map role succession: who fills vacated positions in governance, financial, technical, and protection roles?
- Protection relationship reconstitution: does a disrupted actor acquire a new FSB handler? How quickly? Who is the new officer? Which faction?

Pressure Rotation

- Avoid predictable action patterns: actors adapt to anticipated enforcement rhythms
- Rotate pressure vectors: if financial actions become anticipated, shift to infrastructure or social pressure
- Exploit adaptation: when actors move to new infrastructure or financial rails, those new nodes become targets
- Reassess agency alignment: domestic institutional tolerance shifts over time; re-map quarterly

SECTION 5 | AFFILIATE STRATEGIC FRAMEWORK & RaaS-SPECIFIC DISRUPTION

The affiliate layer is the operational engine of the entire ecosystem. It has historically received less strategic attention than core leadership, a gap this section addresses. The framework below applies to both RaaS franchise operations and closed hierarchical groups.

5.1 Structural Comparison: RaaS vs. Closed Groups

Characteristic	Closed Group (e.g. early Conti, WizardSpider)	RaaS (e.g. LockBit, Qilin, RansomHub)
Structure	Centralized. Employed developers, operators, and negotiators under unified leadership.	Franchise model. Small core team (3-10) provides encryptor, infrastructure, and panel. Affiliates do the hacking.
Revenue split	Centralized revenue; leadership distributes salaries or shares.	70-80% to affiliates, 20-30% admin cut to core team per ransom payment.
Leadership exposure	Higher. Leadership directs operations and is connected to victim activity.	Lower. Core team never touches victim networks. Admin wallet abstracted through multiple layers.
Affiliate role	Employed operators, internal, vetted, salaried or revenue-share.	Independent contractors, external, self-vetted via forum reputation, disposable.
Resilience to takedown	Moderate. Remove leadership, remove the operation.	High. Affiliates migrate to competing RaaS within days. Core team rebrands and re-recruits.
Primary disruption lever	Leadership identification and arrest.	Business model degradation: trust destruction, affiliate risk elevation, cash-out pressure.

5.2 Affiliate Strategic Framework, Both Group Types

Regardless of group structure, the affiliate layer shares common characteristics that make it a high-value strategic target.

Affiliate Prioritization

Priority Tier	Profile	Rationale	Primary Action
Tier 1, High Value	High-volume affiliates responsible for DIB, CIKR, or healthcare targeting; affiliates with documented travel outside Russia/CIS	Maximum operational impact + maximum arrest feasibility	Third-country arrest development; simultaneous financial designation

Priority Tier	Profile	Rationale	Primary Action
Tier 2, Intelligence Value	Affiliates with documented contact methods, forum handles, or infrastructure links to core team leadership	Every affiliate arrest is a potential collection opportunity toward core team identification	Arrest + structured debrief; cultivate as CI with extreme OPSEC
Tier 3, Chilling Effect	High-profile, visible affiliates whose arrest will be observed by the remaining affiliate pool	Arrests chill recruitment more than any single leadership action	Public arrest + maximum public attribution to signal ecosystem-wide risk
Tier 4, Financial Pressure	Affiliates with Western-touchable financial exposure (exchange accounts, real estate, business interests)	Financial pressure without arrest; imposes cost even without custody	OFAC designation + VASP KYC pressure + correspondent banking exposure

Affiliate Mapping Methodology

- Forum vetting history: RaaS affiliates are recruited through Exploit and XSS. Forum registration patterns, vouching relationships, and posting history build identity continuity packages
- Payment address clustering: affiliate payments from ransom proceeds follow predictable patterns, 70-80% splits are detectable in blockchain forensics; affiliate wallet clusters build over time
- Operational signatures: victim targeting patterns, negotiation style, ransom amount calibration, and deployment timing are affiliate-specific fingerprints that persist across group migrations
- Infrastructure reuse: affiliates reuse tooling, staging infrastructure, and C2 patterns across operations and across group migrations, these are identity continuity indicators
- Closed groups: internal communication leaks provide direct affiliate roster visibility; treat every seized or leaked internal dataset as a primary affiliate mapping source

Affiliate Migration Tracking

When a group is disrupted, affiliates migrate. Migration patterns are intelligence, they reveal which competing groups are absorbing talent, what the new operational tempo will be, and who the resilient actors are.

- Pre-position monitoring on likely recipient groups before takedown, not after
- Track forum re-registration and new vouching relationships in the 30 days following disruption
- Map operational signature reappearance: same targeting patterns and deployment timing appearing under a new group banner identifies migrated affiliates
- Migration speed is an ecosystem health indicator: fast migration (under 14 days) indicates high affiliate confidence in alternatives; slow migration indicates ecosystem-level friction

The Affiliate Arrest Multiplier Effect

A single well-chosen affiliate arrest produces effects that extend far beyond the individual:

- Immediate: removes operational capacity, disrupts active victim negotiations
- Intelligence: creates CI pathway toward core team; seized devices yield infrastructure and contact data
- Psychological: remaining affiliates do not know what the arrested affiliate has provided, uncertainty is operationally degrading independent of what was actually shared
- Recruitment: high-profile arrests chill new affiliate recruitment; the RaaS franchise becomes visibly dangerous to join
- Financial: affiliate payment flows seized or frozen impose direct cost on the core team's revenue stream

5.3 RaaS-Specific: Paths to Leadership Identification

Vector	Mechanism	Reliability	Operational Notes
Financial tracing (admin cut)	Admin wallet receives a consistent percentage of every ransom. High-volume recurring flows are harder to fully obscure. Trace through layering, OTC, and exchange withdrawal.	High	Primary path. Requires blockchain forensics combined with exchange KYC pressure. Long timeline but most durable evidence.
Seized infrastructure	Takedown operations on affiliate panels and leak sites yield negotiation logs, affiliate identifiers, payment addresses, and admin access patterns. Operation Cronos (LockBit, Feb 2024) produced direct visibility into admin payment flows.	High (if obtained)	Requires prior takedown. Intelligence from seized panels compounds over time.
Affiliate cooperation	Arrested affiliates know core team contact methods, forum vetting handles, and sometimes infrastructure details.	Medium	Cooperators become CI targets if exposed. Cooperator handling requires extreme OPSEC. Exposure triggers reverse enforcement by Russian agencies.
Developer artifacts in malware	Compile-time metadata, PDB paths, error strings, language settings, and coding style fingerprint individual developers across rebrands.	Medium	Slow but rebrand-resistant. Most valuable for linking a new group to a prior identity after reconstitution.
Underground forum history	RaaS operators maintain reputations on Exploit and XSS. Forum registration patterns, PGP key reuse, and posting style link current identities to prior personas.	Medium	Intel 471 and Flashpoint are primary sources.
Infrastructure persistence	Even OPSEC-conscious groups reuse infrastructure elements	Medium	Document infrastructure fingerprints before takedown

Vector	Mechanism	Reliability	Operational Notes
	across brands: ASN patterns, hosting providers, panel code, domain registration behaviors.		so reconstitution is immediately detectable.

5.4 Attacking the RaaS Business Model

Trust Destruction

- Law enforcement seizure of affiliate panels exposes affiliate identities and active operations, creating fear of exposure across the remaining roster
- Posting apparent evidence of compromise on the group's own infrastructure makes affiliates question whether the platform is burned
- Affiliates who doubt the core team's security migrate to competitors within days, you do not need to arrest the admin to empty the affiliate roster

Payment Rail Pressure

- Designate OTC nodes and exchanges that service RaaS admin wallets
- VASP engagement: flag high-volume criminal clusters for enhanced due diligence at withdrawal points
- Correspondent banking exposure: surface laundering routes touching Western banks
- This is the most durable pressure vector, infrastructure rebuilds in days; financial exposure compounds over time

IAB Layer Disruption

- IAB arrests and underground market disruption reduce quality access supply available to affiliates
- Rising IAB prices compress affiliate margins, reducing franchise attractiveness
- IAB disruption is low-backfire-risk, these actors are not typically state-protected assets

Decryptor Release

- Every publicly released decryptor reduces victim payment rates and undermines core franchise value
- Operation Cronos released 7,000+ LockBit decryption keys publicly
- Even the credible threat of decryptor release reduces victim payment likelihood during active negotiations

5.5 Preventing Reconstitution and Rebrands

- Document infrastructure fingerprints before takedown: panel code, ASN patterns, hosting behaviors, domain registration style
- Maintain malware attribution continuity: code lineage, developer artifacts, crypter relationships

- Immediate public attribution of new brand to prior identity resets brand equity to zero
- Pre-position sanctions designations to fire immediately upon rebrand identification
- If the same developer writes the new encryptor, the same artifacts will appear in samples, malware analysis on day-zero samples closes the attribution gap

5.6 Victim-Side Engagement Framework

The ecosystem has historically been approached from the supply side. Victim payment refusal is one of the most powerful ecosystem pressure mechanisms available, it directly attacks the financial incentive sustaining the entire RaaS model. Every refused payment imposes a cost that no infrastructure rebuild can recover. This section proposes an engagement model, not just a list of mechanisms.

The Engagement Model

Victim-side pressure does not operate through law enforcement alone. It requires coordinated engagement across five institutional channels, each of which owns a different lever. These channels should be engaged in parallel, not sequentially.

Channel	Lever	Ecosystem Effect	Engagement Mechanism
Law Enforcement (FBI/CISA)	Decryptor release + affiliate panel seizure	Direct revenue reduction; undermines franchise value proposition	Coordinate decryptor release with takedown operations; public release maximizes chilling effect on pending victim negotiations
CISA / Sector ISACs	Pre-encryption victim notification	Reduces successful attack completion rate; raises affiliate operational cost	CISA and sector ISACs maintain victim notification pipelines, engage early to ensure high-priority sectors receive alerts before encryption completes
FinCEN / Treasury	Insurer payment policy pressure	Reduces victim payment rate systematically across the ecosystem	FinCEN engagement with cyber insurance sector; ransomware payment coverage restrictions and mandatory law enforcement notification requirements reduce payment rates without legislative action
OFAC	Sanctions payment prohibition	Reduces payment rate for designated groups; compels victim reporting	OFAC designation paired with payment prohibition guidance creates legal liability for victim payment; should be coordinated with takedown timing to maximize disruption
CISA / Sector Regulators	Resilience investment incentives	Long-term structural reduction in victim	Backup and recovery capability eliminates

Channel	Lever	Ecosystem Effect	Engagement Mechanism
		payment rate	payment incentive entirely for resilient victims; resilience standards and investment incentives are the only mechanism that attacks the demand side structurally

Connection to Supply-Side Operations

Victim-side and supply-side operations should be sequenced to compound each other:

- Takedown operations release decryptors → reduces value of pending ransoms → affiliates receive less from ongoing operations → franchise attractiveness declines
- OFAC designations paired with payment prohibition guidance → victim legal liability for payment → payment refusal rate rises → admin cut volume drops → officer protection payment value drops (Section 9 feedback loop)
- CISA pre-notification programs work best when threat intelligence from monitoring operations is shared in time to enable defensive action, this creates a direct operational link between monitoring decisions (Section 6) and victim-side outcomes

► CURRENT EFFECTIVENESS SIGNAL

Ransom payment volumes have declined despite attack frequency increasing. This divergence is the clearest measurable signal that multi-vector ecosystem pressure is producing real compounding friction. The drivers are consistent with the model in this playbook: better victim resilience, law enforcement trust destruction operations, sanctions reducing payment legality, and insurers tightening payment policies.

The critical caveat: attacks staying high while payments fall means groups are working harder for less. Friction is real. But victims are still being hit and data is still being exfiltrated. Payment refusal is a victory; attack prevention is the harder and more important objective.

5.7 HUMINT and Cooperator Handling

The playbook references HUMINT as a source across several sections, particularly for protection relationship reconstitution tracking and core team identification. Given that cooperator exposure triggers active counterintelligence responses by Russian agencies, a brief set of handling principles is warranted. These are offered as analytic input; cooperator programs require specific authority and tradecraft that exceeds this document's scope.

Core Handling Principles

- Treat every cooperating actor as a potentially compromised source from day one. Russian agencies conduct counterintelligence against their own criminal assets. Assume FSB awareness of cooperation is a matter of time, not if.

- Limit knowledge of cooperation status to minimum necessary personnel, in both U.S. and host-country institutions. Leaks that burn arrest windows are well-documented; leaks that burn cooperators carry greater human risk.
- Do not take actions against the cooperating actor's associates that would reveal the source or timing of information. Each action should be justifiable from open-source intelligence alone before the cooperator's information is incorporated into operational decisions.
- Suspected cooperators face serious physical risk. The playbook's engagement trigger table (Section 8) identifies cooperator exposure as activating reverse enforcement by Russian agencies. Design debrief programs and subsequent operations with this explicitly in mind.
- Structure debriefs to systematically extract officer contact methods, protection payment documentation, and infrastructure access credentials, the protection layer track depends on this data and it is rarely available through OSINT alone.
- Do not terminate cooperation abruptly. Use decay rather than cutoff to reduce exposure risk, declining contact frequency over time is less detectable than a clean break following an operational action.

Any cooperator program requires dedicated tradecraft expertise and appropriate authority beyond this document's scope.

SECTION 6 | TAKEDOWN vs. MONITORING: THE CORE OPERATIONAL DECISION

Whether to take down infrastructure or continue monitoring it is one of the most consequential decisions in disruption operations. The central error is treating them as competing options. They are sequential phases of a single operation.

6.1 The Core Tension

Takedown produces immediate disruption and, if infrastructure is seized rather than just shut down, an intelligence windfall, potentially years of logs, user data, transaction records, and admin access patterns. Monitoring produces ongoing intelligence but allows real harm to continue while you watch.

Monitoring advantages: reveals operational intent, planned attacks, internal disputes, and full network structure before disruption. The panic signal after a takedown, who disappears, who migrates, who tries to contact whom, is itself intelligence on network structure. Live channels with operational planning visible generally outweigh the disruption value of shutting them down, unless imminent victim harm is preventable.

Takedown advantages: burn risk, if your access is discovered, you lose the intelligence and actors migrate to a hardened platform. Seized infrastructure often exceeds the value of continued monitoring. The seizure itself is a trust destruction weapon: remaining actors do not know who cooperated or what law enforcement now knows.

6.2 Decision Threshold: When to Move from Monitor to Takedown

Trigger	Rationale	Risk if Delayed
Ongoing victim harm exceeds intelligence value	Particularly if critical infrastructure, healthcare, or DIB targets are being hit. Continued monitoring becomes legally and ethically indefensible.	Legal/oversight exposure; reputational damage if monitoring is disclosed
Monitoring access is at risk of discovery	Sophisticated actors conduct counterintelligence. A controlled takedown on your terms is better than a discovered access.	Loss of all intelligence; actors harden new platform
Sufficient ecosystem mapping to support follow-on actions	Leadership identified or substantially narrowed; affiliate roster mapped; financial flows traced. Marginal value of additional monitoring is declining.	Diminishing returns; unnecessary harm continuation
Time-sensitive operational opportunity	A key actor is traveling outside Russia, a financial window exists for simultaneous designations, or a partner operation creates a coordination moment.	Missed arrest or designation window
Coordinated multi-partner action is ready	Maximum disruption requires simultaneous action across jurisdictions.	Partner readiness degrades; coordination gaps widen

Trigger	Rationale	Risk if Delayed
	When all partners are aligned, delay reduces coordination quality.	

► **BOTTOM LINE**

Monitor to map. Take down at maximum yield. Use seized data to pursue follow-on actions. Monitor reconstitution to target the next iteration. The takedown is a phase, not an endpoint.

SECTION 7 | RUSSIAN AGENCY QUICK REFERENCE

Confidence levels reflect observed enforcement behavior, not legal doctrine. Note: FSB is not treated as a unified actor. Internal factions, competing officer interests, and political dynamics within FSB create divergent enforcement behaviors, the column below reflects FSB's aggregate behavior while Section 9 operationalizes the factional divergences.

Agency	Role in Ecosystem	Best Leverage Points	Confidence	Key Limits
FSB	Primary architect of cyber ecosystem; recruits, co-opts, or protects actors for CI and strategic ops. Internally factional, competing officers and units have divergent interests that can be exploited.	Attribution fallout; actor defiance of recruitment; actor ties to foreign intelligence; manufactured liability via protection layer exposure (Section 9); factional competition between FSB units	High (selective)	No legal cooperation as an institution. Foreign pressure reduces enforcement appetite at agency level. Individual factions may act when actor defies, embarrasses, or becomes a liability to specific officers or units.
GRU	Leverages actors for military/geopolitical ops; rarely arrests but will silence or cut off. Note: actors absorbed into wartime GRU operations fall outside this framework's scope.	Sloppy OPSEC; attribution risk to ongoing ops; actor disobedience post-campaign	Low	No transparency; no prosecutorial handoff. Disruption must be indirect.
MVD / Dept K	Enforces mid-level fraud and technical cybercrime; reputationally sensitive; sidelined in elite cases	Domestic financial harm; media scrutiny; interagency competition with FSB; arrest metrics pressure	High (non-RIS)	FSB can override at any time. Not accessible for RIS-linked actors.
FNS	Identifies shell companies, undeclared income, and lifestyle inconsistencies; no arrest power	Financial irregularities; family asset exposure; laundering front structures	Med-High	Action requires political greenlight. Surfaces exposure; does not prosecute.
Rosfinmonitoring	Russia's financial intelligence hub; maps laundering infrastructure; triggers asset controls and referrals	Crypto-fiat flows; suspicious transactions; Egmont Group scrutiny pathways	High (mapping)	Politicized. Requires downstream adoption by MVD or FSB to produce arrests.
CBR	Via 115-FZ, enables	Suspicious	Med-High	Does not attribute

Agency	Role in Ecosystem	Best Leverage Points	Confidence	Key Limits
	banks to freeze or deny transactions based on risk without prosecution	transaction patterns; politically exposed clients; AML risk flags		activity or target individuals. Non-attributable friction only.
SKR	Engages only when cybercrime is elevated to elite criminality or political scandal	Political embarrassment; regime exposure; organized crime framing (Art. 210)	Medium	Not reachable through standard LE channels. Requires Kremlin-level political signal.
Roskomnadzor	Restricts communications platforms; imposes infrastructure friction	Hosting noncompliance; foreign platform resistance; digital sovereignty framing	Medium	Does not target individuals. Disrupts legitimate users equally.

SECTION 8 | ENGAGEMENT TRIGGERS TO AVOID

These patterns consistently cause Russian agencies to protect, absorb, or redirect cybercriminals rather than suppress them. They are drawn from historical enforcement outcomes and institutional behavior analysis. Red = high backfire risk (avoid or delay until actor is isolated). Amber = conditional risk (proceed with domestic framing in place).

Trigger	Effect	Mechanism	Implication for Operations
Public attribution by foreign government	Actor converts from criminal liability to national security asset	Once named by a foreign power (especially the U.S.), FSB/GRU may treat actor as soft-state asset regardless of prior behavior	Attribution hardens protection. Publicity equals absorption. Delay public attribution until actor is already isolated.
Formal arrest or extradition request	Triggers defensive nationalism; reduces arrest probability to near zero	Russian doctrine opposes surrender of nationals. Extradition requests signal foreign ownership of the case.	Extradition-first strategies produce the opposite of suppression. Pursue third-country arrest opportunities instead.
Media naming and shaming campaigns	Agencies treat engagement as hostile information warfare; enforcement appetite declines	When actors are labeled 'Russian cybercriminals' without domestic impact framing, it reads as sovereignty violation	Paired domestic framing is required before any public naming.
Indication of actor cooperation with foreign LE	Actor becomes a counterintelligence interest; FSB views as double-agent risk	Suspected cooperators are arrested, disappeared, or neutralized	Cooperator handling requires extreme operational security. Exposure of cooperation triggers reverse enforcement.
Actor technical value or recruitment potential	Delays or cancels enforcement; actor becomes reusable state asset	Actors with malware or infrastructure capabilities are considered recruitable	Prioritize disruption of capability before it triggers recruitment.
Target selection aligned with Russian strategic interests	Actor becomes functionally aligned with state objectives; impunity follows	Operations against Western banks, NATO infrastructure, or sanctions enforcement are considered symbiotic by Russian state	Document target patterns to predict and preempt state absorption.
Internal elite sponsorship (krysha)	Immunity from arrest regardless of cybercrime visibility	Actor operates under protection of regional, political, or RIS patron	Pressure must first weaken or circumvent sponsor relationship. FNS/Rosfinmonitoring exposure of patron is prerequisite.

Trigger	Effect	Mechanism	Implication for Operations
Multilateral pressure without local framing	Resistance from all Russian agencies; interpreted as sovereignty violation	Pressure through Western consortiums without Russian criminal charge equivalents reads as hostile	Align multilateral pressure with simultaneous domestic framing.

► BOTTOM LINE

Russian agencies protect cybercriminals when engagement signals foreign ownership, regime threat, or operational opportunity. Effective disruption depends on avoiding these triggers while exploiting internal contradictions and institutional sensitivities.

SECTION 9 | PROTECTION LAYER DISRUPTION: FSB OFFICER LIABILITY TRACK

9.1 Core Concept: Protection as a Load-Bearing Ecosystem Node

FSB officers who provide krysha, recruitment, and protection to cybercriminal actors are not peripheral to the ecosystem, they are load-bearing nodes. Disrupting a criminal actor whose protection relationship remains intact produces reconstitution. Disrupting the protection relationship first, or simultaneously, produces ecosystem degradation.

The mechanism: individual FSB officers maintain protection relationships because they generate personal value, financial, operational, reputational within the service. When a specific protection relationship generates more cost than value, domestic embarrassment, attribution risk to the officer personally, institutional liability to FSB leadership or competing factions, the pressures on that officer change. You are not asking FSB to cooperate as an institution. You are creating conditions where a specific officer's calculus shifts, or where competing FSB factions find it advantageous to act against the liability-generating officer. These are different and more achievable objectives.

This distinction matters operationally. FSB's internal factions do not share interests. An officer in the FSB's economic security directorate and an officer in a cyber unit may be in active competition. Surfacing one officer's criminal financial exposure to competing FSB units, rather than to FSB 'leadership' as an abstraction, is a more precise and realistic targeting objective. Confidence: Medium-High (mechanism is well-supported; execution difficulty is high)

Worked example (Karakurt / Zolotarjovs, May 2026). On 4 May 2026 a US court sentenced Deniss Zolotarjovs to 102 months for his role in Karakurt, a data-extortion group the Department of Justice describes as led by former Conti leadership. DOJ stated the group relied on access to Russian government databases and law-enforcement connections, and paid bribes to secure military draft exemptions and tax avoidance for its members. This is the clearest open-source confirmation to date of the premise in this section: protection is a purchased, load-bearing relationship with named state touchpoints (government databases, draft boards, tax authorities), not a vague tolerance. It also illustrates the officer-liability logic in reverse. The same corruption that shields members is itself documentable financial exposure, exactly the paper trail the financial-exposure methodology below is built to surface. Confidence: Confirmed (DOJ statement and sentencing, 4 May 2026).

9.2 Officer Relationship Types and Leverage

Officer Relationship Type	Protection Mechanism	Liability Trigger	Primary Lever
Direct handler / recruiter	Tasks criminal actors for intelligence collection or strategic ops; provides operational cover	Sloppy OPSEC by the criminal actor creating attribution risk back to the officer; criminal actor creating domestic harm visible enough to attract MVD attention	Attribute criminal actor's domestic harm to officer's operational portfolio; surface attribution risk to competing FSB units

Officer Relationship Type	Protection Mechanism	Liability Trigger	Primary Lever
Krysha / protection provider	Financial relationship, officer receives payment to ensure enforcement non-interference	Financial exposure: payment flows documented and surfaced domestically; Western asset exposure via sanctions	Anomalous outflow analysis to document payment relationship; FNS lifestyle referral on officer; simultaneous criminal network + officer sanctions
Passive tolerance / non-interference	Officer aware of actor but chooses not to act; implicit protection through inaction	Actor creates embarrassment or domestic harm sufficient to make the officer's inaction visible to superiors or competing units	Domestic harm amplification via investigative journalism; FNS lifestyle flags on actor to create paper trail officer cannot ignore
Retired / former officer	Residual institutional relationships and access used to provide informal protection	No active institutional protection; most vulnerable to financial sanctions and third-country arrest	Western asset designation; third-country arrest development (Annex A); family financial exposure

9.3 Financial Exposure Methodology for Officers

FSB officers on government salary who maintain protection relationships with high-volume cybercriminal actors accumulate financial exposure that does not match their declared income. Building this exposure profile requires no HUMINT, it is an open source and financial registry analytic task that extends directly from the criminal-side financial mapping already in this playbook. All deliverables in this section are proposed; none currently exist.

Step 1: Identify Anomalous Outflows from Criminal Actor Finances

This is the keystone analytic task. Detailed methodology in Section 10. For this section: the question to ask of every criminal actor's financial map is what outflows do not fit criminal operational cost profiles.

- Criminal operational costs: hosting, tooling, affiliate payments, mule recruitment, OTC fees, developer salaries, all have identifiable patterns and counterparties
- Protection payments sit outside these patterns: they are recurring, consistent in amount or proportion, do not flow to known criminal operational nodes, and often convert to fiat through channels separate from the main cash-out infrastructure
- Flag these anomalous outflows as protection payment candidates, they are the entry point to the officer financial profile

Step 2: Build the Officer Financial Profile

- Russian property registries: egrn.ru and regional equivalents show real estate holdings in officer and immediate family names
- Corporate registries: egrul.nalog.ru surfaces business ownership, shell companies, consulting firms, and holding structures in spouse or children's names
- European property and corporate registries: country-specific registries (see Annex A) surface Western-held assets; Cyprus, Malta, and pre-2022 EU jurisdictions have historically been favored
- Travel and lifestyle signals: visa application histories, social media, school enrollment records for children, vehicle registrations, build lifestyle inconsistency profile against declared government salary
- Cross-reference: connect anomalous criminal outflows to officer financial profile via timing, amount consistency, and geographic correlation of cash-out endpoints

Step 3: Channel Selection for Exposure

Channel	Mechanism	Backfire Risk	Best Use Case
FNS referral (domestic)	Lifestyle inconsistency and undeclared income flagged to Federal Tax Service, no foreign fingerprint, purely domestic process	Low	Officers with documented Russian-held assets inconsistent with salary; generates domestic paper trail
Rosfinmonitoring referral	Suspicious transaction flags on protection payment flows, feeds CBR 115-FZ freeze pipeline	Low	High-volume payment flows between criminal actor and officer-linked accounts
Investigative journalism (OCCRP / Bellingcat / iStories / Meduza)	Financial and lifestyle documentation provided to investigative outlets, produces domestic scandal framing rather than foreign attribution	Medium (lower than official attribution)	Officers with Western asset exposure and documentable lifestyle inconsistency; creates domestic embarrassment without official foreign-government fingerprint
Simultaneous OFAC/OFSI designation	Officer + criminal network designated simultaneously; Western asset freeze + correspondent banking pressure	Medium-High (acceptable if domestic framing pre-positioned)	Retired/former officers or officers already domestically exposed; batch with criminal network to maximize impact
Third-country legal pre-positioning	Sealed indictments or arrest warrants filed in viable European jurisdictions; activated when travel window opens	Low (if kept sealed)	Officers with documented travel patterns to viable jurisdictions (Annex A)

9.4 Manufacturing Internal FSB Contradictions

The goal is not to get FSB to cooperate as an institution, it is to make specific officers liabilities that competing FSB units or FSB leadership's factional interests will act against. This requires surfacing the right information to the right internal FSB audience, which requires knowing which FSB factions are in competition with the officer's unit. Confidence: Medium (mechanism is sound; intelligence requirements are high)

What Makes an Officer a Liability to FSB Leadership or Competing Factions

- Attribution risk: the officer's criminal protection relationship creates a documented link between a specific FSB unit and a ransomware operation that has hit high-profile targets , particularly U.S. critical infrastructure or healthcare, giving competing FSB units leverage against that officer's unit
- Domestic embarrassment: the officer's financial exposure becomes visible in Russian domestic media or through FNS/Rosfinmonitoring processes, creates pressure on leadership to demonstrate institutional integrity
- Operational incompetence signal: the protected actor gets disrupted repeatedly despite protection, makes the officer look ineffective and the protection relationship a waste of institutional resources
- Foreign entanglement risk: evidence that the officer's financial exposure involves Western assets creates counterintelligence concern for FSB leadership, is this officer a recruitment target for foreign intelligence services? This framing activates FSB's counterintelligence units as a potential instrument against the officer

9.5 Portfolio Sequencing for Officer Targets

Tier	Officer Profile	Action Sequence	Timeline
Tier 1, Immediate	Retired/former officers with Western asset exposure and active criminal network ties	Financial profile build → OFAC/OFSI designation (officer + criminal network simultaneously) → third-country arrest legal pre-positioning → investigative journalism pipeline	0-90 days
Tier 2, Financial Exposure	Active duty krysha officers with documentable payment relationships	Anomalous outflow identification → FNS referral → Rosfinmonitoring flag → investigative journalism (if profile is strong) → OFAC designation after domestic exposure is established	90-180 days
Tier 3, Contradiction Manufacturing	Active duty handlers whose protected actors are generating domestic harm or attribution risk	Domestic harm amplification → attribution risk surfacing to competing FSB unit awareness → operational incompetence signaling via repeated disruption of protected actor → monitor for protection relationship strain	180-365 days
Tier 4, Long Lead	Active duty officers with unclear status or current high collection value	Monitor only, define trigger conditions before monitoring begins; do not act until protection relationship mapping is complete and successor officer is identified	Ongoing

► COMPOUNDING FEEDBACK LOOP

Criminal-side financial pressure → anomalous outflow identification → officer financial profile → FNS/investigative journalism exposure → domestic liability for officer → FSB factional pressure on officer's relationship → criminal actor loses protection → criminal actor becomes accessible to MVD enforcement → MVD enforcement generates more intelligence on financial flows → stronger anomalous outflow identification → repeat. Each cycle strengthens the next.

SECTION 10 | CRYPTO-TO-FIAT METHODOLOGY

10.1 Why This Is the Keystone Analytic Task

Every financial pressure action in this playbook, wallet designations, VASP engagement, correspondent banking exposure, OTC node targeting, protection payment identification, requires knowing where funds become fiat and who touches them. Without a documented cash-out graph, financial pressure actions are targeted at symptoms rather than structural nodes. With it, single actions produce cascading disruptions across multiple actor flows.

The cash-out graph also contains the protection payment data that drives the Section 9 officer liability track. Anomalous outflows, payments that do not fit criminal operational cost profiles, are only identifiable against a complete model of what criminal operational costs look like. Build the model first.

10.2 The Cash-Out Graph: Layer Structure

Layer	What It Contains	Key Questions	Primary Sources
Layer 1: Ransom Receipt	Initial ransom payment wallets; victim-to-actor payment flows; multi-sig escrow structures used in negotiation	What wallets receive ransom payments? Are they reused or single-use? What mixing or structuring occurs immediately post-receipt?	Chainalysis Reactor; TRM Labs; on-chain forensics
Layer 2: Layering	Mixing services, chain-hopping (BTC → Monero → BTC), structuring into sub-threshold amounts, peel chains, consolidation wallets	How many hops before funds reach a cash-out node? What mixing services are used? Are layer 2 wallets shared across multiple actor flows?	Chainalysis; TRM; Elliptic, cross-validate outputs
Layer 3: Pre-Cash-Out Aggregation	Consolidation wallets that aggregate layered funds before exchange deposit or OTC transfer; admin cut separation at this layer	Where does the admin cut separate from affiliate payments? What wallets aggregate funds from multiple ransom flows? These are high-value designation targets.	Blockchain forensics, look for consistent percentage splits
Layer 4: Cash-Out Nodes	OTC brokers, exchange deposits (VASP), peer-to-peer platforms, payment aggregators, crypto ATMs	Which specific OTC nodes and exchange accounts receive funds? What are the withdrawal patterns? What geographic concentration exists?	Chainalysis + exchange KYC pressure; Intel 471 for OTC broker identification
Layer 5: Fiat Entry	Bank accounts, payment systems, real estate purchases, front company revenue, luxury asset acquisition	Which banks receive fiat proceeds? What front companies hold proceeds? Where does money enter the legitimate financial	Rosfinmonitoring referrals; FNS corporate registry; property registries; correspondent banking data

Layer	What It Contains	Key Questions	Primary Sources
		system?	

10.3 Anomalous Outflow Identification

Once the standard cash-out graph is built, anomalous outflows become visible as flows that do not fit the expected pattern at each layer. These are protection payment candidates.

Criminal Operational Cost Baseline

Establish what normal operational costs look like before flagging anomalies:

- Hosting and infrastructure: BPH provider payments, domain registration, server rental, typically small, recurring, technically-attributable
- Affiliate payments: 70-80% of ransom value flowing to affiliate wallets, consistent split ratio, identifiable at Layer 3
- Developer and tooling costs: malware developer payments, crypter services, IAB access purchases, typically one-time or irregular, flow to identifiable underground market nodes
- Mule recruitment and cash-out fees: OTC broker fees (typically 1-5%), mule handler payments, identifiable by counterparty type
- Operational security costs: VPN services, infrastructure rotation, OPSEC tooling, small and recurring

Anomalous Outflow Indicators

- Consistent percentage flows to wallets with no identifiable criminal operational counterparty
- Recurring payments of consistent amounts on regular intervals, salary or retainer patterns not consistent with contractor relationships
- Flows that convert to fiat through separate channels from main cash-out infrastructure, protection payments are often kept operationally separate from the main financial flow
- Payments that predate or follow enforcement non-action on a specific actor, temporal correlation with protection delivery is a strong indicator
- Flows to wallets that connect to Russian domestic banking through politically-exposed-person adjacent accounts

► ANALYTICAL DISCIPLINE NOTE

Anomalous outflow identification requires a complete operational cost baseline before anomalies can be flagged. Designating a wallet as a protection payment candidate without establishing what normal looks like produces false positives that undermine subsequent legal action. Build the baseline first. Flag anomalies against it. Cross-validate across multiple ransom payment flows for the same actor before drawing conclusions.

10.4 OTC Broker Network Mapping

OTC brokers are the most critical cash-out bottleneck in the Russian ransomware ecosystem. Unlike exchanges, they are relationship-based, less regulated, and serve as the primary bridge between crypto and Russian domestic fiat for high-volume criminal actors.

OTC Identification Methodology

- Blockchain cluster analysis: OTC brokers accumulate funds from multiple unrelated criminal sources, their wallets show a distinctive pattern of diverse inflows and consistent outflow to banking channels
- Underground forum advertising: major OTC brokers advertise on Exploit, XSS, and Telegram channels, Intel 471 and Flashpoint provide primary visibility
- Exchange KYC linkage: OTC brokers often maintain exchange accounts for liquidity, KYC pressure on exchanges surfaces OTC broker identities
- Rosfinmonitoring cross-reference: high-volume fiat conversion through Russian banking channels without obvious legitimate business origin is flagged by Rosfinmonitoring, cross-reference with blockchain forensics to identify the crypto side of the same broker

OTC Network Graph Components

- Tier 1 brokers: high-volume, serve multiple ransomware groups simultaneously, these are ecosystem-critical nodes whose designation produces cascading effects across multiple actor financial flows
- Tier 2 brokers: group-specific or actor-specific; less systemic impact but more directly traceable to specific actors
- Mule network integration: OTC brokers frequently operate mule recruitment networks as a complementary service, map the connection between OTC operations and mule handler networks
- Geographic concentration: Russian OTC operations tend to concentrate in Moscow, Saint Petersburg, and Yekaterinburg, geographic mapping supports physical surveillance and arrest operations

SECTION 11 | MEASUREMENT FRAMEWORK

Disruption operations without measurement produce narrative, not accountability. This section defines the minimum viable measurement posture for tracking ecosystem health and operational effectiveness. None of these measurement capabilities currently exist; they are proposed for build-out.

11.1 Establishing Baselines

KPIs defined without documented baselines cannot be used to attribute change to specific actions versus ecosystem-level trends. Establish baselines before any pressure actions are taken.

- For each KPI: document the 6-month rolling average at the time operations commence, this is your pre-intervention baseline
- Establish a control period of 60-90 days before first pressure action during which KPIs are tracked but no actions taken, this separates pre-existing trend from intervention effect
- Document all significant external events (major law enforcement actions by other agencies, geopolitical developments, cryptocurrency market movements) that could independently affect KPIs, these are confounding variables that must be noted when interpreting results
- For the officer liability track: establish a baseline protection relationship map (Section 9 mapping) before any exposure operations, changes in protection status are only measurable against a documented prior state

11.2 Ecosystem Health KPIs

KPI	What It Measures	Source	Cadence
Active ransomware groups (count)	Total groups posting victims; proxy for ecosystem breadth	Ransomware.live	Monthly
Victims posted per week (rolling average)	Operational tempo; leading indicator of ecosystem health	Ransomware.live	Weekly
Ransom payment volume (USD)	Financial incentive sustaining ecosystem; lagging indicator	Chainalysis	Quarterly
Time to reconstitution after takedown	Resilience measure; declining trend = increasing friction	Censys / internal	Per event
BPH provider count (active)	Infrastructure supply; declining trend = upstream pressure working	Shadowserver / Censys	Monthly
Sanctioned wallet activity (post-designation)	Sanctions effectiveness; continued activity = compliance gap	Chainalysis / TRM	Monthly
Underground market activity index	Forum post volume, pricing stability, trust-node activity; proxy for market health	Intel 471 / Flashpoint	Quarterly

KPI	What It Measures	Source	Cadence
New IAB listings (count)	Supply of network access available to ransomware affiliates	Intel 471 / Flashpoint	Monthly
Affiliate migration speed post-disruption	Ecosystem resilience; fast migration = low friction; slow = increasing cost	Intel 471 / forum monitoring	Per event
Victim payment refusal rate (where measurable)	Victim-side resilience; increasing trend compounds supply-side pressure	Coveware / insurer reporting	Quarterly
Protection relationship reconstitution time	How quickly disrupted actors acquire new FSB protection; declining speed = officer network under pressure	HUMINT / forum monitoring / enforcement gap analysis	Per event

11.3 Protection Layer Leading Indicators

Reconstitution time is a lagging indicator, it measures outcomes after the fact. The following leading indicators are proposed to track Section 9 progress before protection relationship changes are observable in reconstitution data:

Leading Indicator	What It Signals	Source	Cadence
Officer lifestyle exposure pipeline activity	FNS referrals filed; investigative journalism materials prepared and delivered, signals that exposure operations are in motion before domestic effects are visible	Internal tracking	Monthly
Anomalous outflow identification progress	Number of protection payment candidates documented with cross-validation, leading indicator for both officer profiles and designation packages	Internal tracking	Monthly
Criminal actor forum signals re: protection confidence	Underground forum discussions showing actor uncertainty about protection status, complaints about krysha quality, or explicit concern about handler reliability, these precede protection relationship breakdown	Intel 471 / Flashpoint / forum monitoring	Monthly
Reconstitution attempts without normal protection speed	Actor attempts to reconstitute more slowly than historical baseline, or without the	Censys / internal monitoring	Per event

Leading Indicator	What It Signals	Source	Cadence
	infrastructure access that protected actors typically have, signals protection may be weakening before explicit breakdown		
Protection payment flow routing changes	Anomalous outflows rerouting, declining in volume, or converting through different channels than established pattern, may signal actor concern about protection relationship integrity	Chainalysis / TRM	Monthly

11.4 Pressure-Effect Ledger

Every significant pressure action should be logged in a structured ledger. This enables retrospective analysis of what worked, what failed, and what adaptation patterns emerged.

Field	Description	Required?
Action date	Date action was taken	Yes
Action type	Sanction / takedown / referral / designation / exposure / officer liability action / other	Yes
Target	Actor, wallet, domain, provider, node, or officer targeted	Yes
Expected effect	What outcome was predicted and over what timeframe	Yes
Observed effect	What actually happened; include null result if no observable change	Yes
Time to reconstitution	Days until actor resumed operations or infrastructure re-appeared	If applicable
Protection status change	Did protection relationship change following action? New officer? Weakened protection? New FSB faction involved?	If applicable
Adaptation pattern	How actor adapted; use this to update substitutability models	If applicable

11.5 Recommended Investment Priorities

These recommendations are offered for consideration by operational and policy stakeholders. None are currently resourced; all require appropriate authority and partner coordination before implementation.

- Intermediary cash-out mapping: converting laundering concepts into repeatable, case-linked cash-out graphs (Section 10). This single investment raises financial plumbing coverage more than any other action. Proposed minimum deliverable: top 20 OTC/broker

nodes + mule recruitment patterns + 3-5 end-to-end case exemplars + anomalous outflow registry.

- Protection layer financial exposure capability: applying the Section 10 methodology specifically to identify anomalous outflows as protection payment candidates, then building officer financial profiles using open source registry analysis. Proposed minimum deliverable: anomalous outflow registry for top 20 protected actors + officer financial profiles for identified protection relationships.
- Affiliate mapping and succession pre-positioning: building full affiliate rosters for priority groups with substitutability assessments and pre-positioned pressure on identified successors. Proposed minimum deliverable: affiliate roster + operational signature database + successor map with pre-positioned pressure packages for top 5 groups.
- Underground governance and trust node mapping: identifying escrow operators, forum administrators, and arbitrators, non-technical but ecosystem-critical, low-backfire-risk targets. Proposed minimum deliverable: top 10 trust nodes + dependency analysis + escrow/arbitration record exploitation plan.
- Measurement layer build-out: implementing baseline documentation, KPI set, and pressure-effect ledger. Without this, effectiveness claims rest on narrative alone. Proposed minimum deliverable: KPI dashboard (monthly cadence) + baseline documentation + logging template tied to every major action. See companion document: Ransomware Ecosystem Disruption Measurement Framework (KPI Architecture) for the complete measurement architecture, confidence labeling system, and per-operation log template.
- Identity and continuity package: building cross-platform handle mapping, PGP reuse tracking, panel fingerprints, and brand-asset reuse patterns. Proposed minimum deliverable: cross-platform handle map + PGP reuse database + brand-asset reuse patterns for top 10 groups.

SECTION 12 | TOOL & PARTNER REFERENCE

Vendors provide validation, not conclusions. Outputs should be cross-checked across multiple sources before driving operational decisions.

Tool / Partner	Category	When to Use	Key Questions to Ask
Chainalysis Reactor / Data Solutions	Blockchain attribution	Wallet clustering, sanctions exposure screening, laundering typology mapping, tracing from ransom payment to cash-out	What assumptions underpin the clustering? Where does attribution confidence drop? What exchanges or OTC nodes touch the end of the chain?
TRM Labs / Elliptic	Blockchain attribution	Cross-validation of Chainalysis outputs; sanctions screening; VASP risk profiling	Where do outputs diverge from Chainalysis, and why? What is the confidence basis for VASP compliance risk scoring?
Mandiant / CrowdStrike Intelligence	Malware & campaign intel	Malware lineage and evolution, affiliate migration, tradecraft shifts, actor attribution	What would falsify this attribution? What evidence supports continuity vs. rebrand? What replaces this toolchain within 30 days?
ESET Research / Kaspersky (open- source only)	Malware & campaign intel	Technical malware analysis, campaign tracking; Kaspersky limited to public reporting with verification	What is the publication basis? For Kaspersky: has this been corroborated by a second source?
Intel 471 / Flashpoint	Underground monitoring	Underground market dynamics, pricing, trust relationships, actor reputation, recruitment channels, forum activity	What is the source methodology? How current is the access? Where does underground visibility drop off?
Ransomware.live / RansomLook	Victim & campaign tracking	Real-time victim counts, leak site monitoring, group activity tracking, ecosystem health KPIs	What is the data lag? Are posting dates confirmed or estimated? What groups are absent from monitoring?
Shadowserver / Censys	Infrastructure attribution	Infrastructure persistence and reconstitution patterns, ASN and hosting clustering, BPH provider mapping	What upstream dependencies are shared across multiple criminal nodes? How quickly does reconstitution appear in scan data?
Recorded Future / Microsoft MDTI	Infrastructure & OSINT	Domain and IP intelligence, cross- platform OSINT fusion, threat actor profiling, infrastructure persistence	What is the evidence basis for actor-to-infrastructure attribution? What is the confidence tier?

Tool / Partner	Category	When to Use	Key Questions to Ask
OCCRP / Bellingcat / iStories / Meduza	Investigative journalism pipeline	Officer financial exposure and lifestyle documentation; surfacing protection relationship evidence through non-attributable channels	Is the documentation package sufficient to withstand investigative scrutiny? Does it contain USG fingerprints that would trigger backfire? Is domestic framing pre-positioned?

ANNEX A | EUROPEAN JURISDICTIONAL FRAMEWORK

This annex provides country-by-country assessment of European jurisdictions for third-country arrest viability, legal pre-positioning requirements, and treaty landscape for Russia/CIS-linked cybercriminal actors and FSB officers. Five Eyes jurisdictions (US, UK, Canada, Australia, New Zealand) are treated as assumed known baseline and not covered here.

Assessment criteria for each jurisdiction: extradition treaty status with the U.S.; treaty status with Russia (a bilateral extradition treaty with Russia makes the jurisdiction less viable as Russia can request competing extradition); rule of law and judicial independence; historical cooperation on cybercrime cases; travel pattern intelligence; and practical arrest infrastructure (liaison relationships, legal pre-positioning lead time).

A.1 Tier 1, High Viability (Priority Pre-Positioning)

Germany

Extradition treaty (U.S.)	Yes, bilateral treaty; active cooperation history
Treaty with Russia	None, Russia cannot compete for extradition
Cooperation track record	High, Operation Endgame (2024) demonstrated deep BKA/FBI/Europol cooperation; German prosecutors have filed independent cybercrime indictments
Travel pattern relevance	Significant Russian business and diaspora community; transit hub for Eastern European actors traveling west
Pre-positioning lead time	60-90 days for coordination with BKA and German federal prosecutors; MLAT requests processed efficiently
Key limits	German courts require substantial evidentiary basis before issuing arrest warrants on foreign requests; political sensitivity around Russia-related cases post-2022 has increased rather than decreased cooperation willingness
Verdict	VIABLE, TIER 1. Priority pre-positioning jurisdiction. BKA relationship and Operation Endgame precedent make this the strongest European arrest jurisdiction for cybercrime.

Netherlands

Extradition treaty (U.S.)	Yes, active cooperation; NHTCU has deep FBI/DOJ relationship
Treaty with Russia	None
Cooperation track record	Exceptional, Operation Cronos (LockBit), Hive takedown, DoubleVPN, RaidForums all involved Dutch jurisdiction; NHTCU is among the most capable and cooperative cybercrime units in Europe
Travel pattern relevance	Amsterdam Schiphol is a major transit hub; significant Russian business presence; financial sector attracts criminal financial infrastructure

Pre-positioning lead time	45-60 days; established MLAT channels and existing working relationships accelerate pre-positioning
Key limits	Dutch legal standards for provisional arrest require imminent flight risk documentation; judges are independent and will scrutinize evidentiary basis
Verdict	VIABLE, TIER 1. Possibly the single strongest European jurisdiction for cybercrime arrests. Default first-choice pre-positioning jurisdiction.

Spain

Extradition treaty (U.S.)	Yes, bilateral treaty; multiple successful extraditions
Treaty with Russia	None
Cooperation track record	Good, multiple Russia/CIS cybercrime arrests; Spanish law enforcement has demonstrated willingness to act on U.S. requests
Travel pattern relevance	High, favored destination for Russian/CIS criminal actors and oligarchs; Costa del Sol and Barcelona have significant Russian community presence; known residence jurisdiction for multiple cybercriminal actors
Pre-positioning lead time	60-90 days; Guardia Civil and Policia Nacional have established FBI liaison relationships
Key limits	Spanish judicial process can be slow; provisional arrest requests require prompt follow-up with formal extradition documentation or the subject must be released
Verdict	VIABLE, TIER 1. High travel pattern relevance elevates Spain as a high-priority pre-positioning jurisdiction, particularly for actors known to reside or vacation there.

A.2 Tier 2, Conditional Viability (Compressed Assessment)

Jurisdiction	Key Strengths	Key Limits	Verdict
France	Bilateral U.S. extradition treaty; Paris and Riviera have Russian high-net-worth presence; ANSSI and DGSI have participated in joint operations	Judicial process is slow (90-120 day pre-positioning lead time); French courts are genuinely independent, provisional arrest without strong evidentiary package risks release	TIER 2, CONDITIONAL. Secondary option; do not rely as primary jurisdiction unless actor has documented presence.
Poland	Bilateral U.S. extradition treaty; no Russia extradition treaty; strong post-2022 political motivation to counter Russian-linked activity; ABW expanding cybercrime cooperation	Less established MLAT infrastructure; judicial standards less predictable than Western European counterparts; more relevant for Eastern European actors than Russian actors	TIER 2, CONDITIONAL. Increasing viability post-2022. Best for Eastern European-based actors transiting Poland.

Jurisdiction	Key Strengths	Key Limits	Verdict
		specifically	
Czech Republic	Bilateral U.S. extradition treaty; no Russia treaty; Nikulin arrest (2017) proved viability; Czech courts withstood Russian diplomatic pressure	Nikulin-style cases attract significant Russian diplomatic pressure; Czech authorities held firm but the political cost was real	TIER 2, CONDITIONAL. Proven jurisdiction with established precedent. Russia will apply maximum diplomatic pressure on high-profile cases.
Italy	Bilateral U.S. extradition treaty; no Russia treaty; high Russian high-net-worth residential presence (northern Italy, Sardinia)	Slow judicial process (90-120 day lead time); inconsistent cooperation track record; provisional arrest procedures less streamlined than Northern Europe	TIER 2, CONDITIONAL. High travel relevance but slower and less predictable. Pre-position for actors with documented Italian presence; not primary for time-sensitive operations.
Greece	Bilateral U.S. extradition treaty; no Russia treaty; Vinnik arrest (2017) demonstrated viability; high Russian tourist and transit traffic; significant Russian property ownership	Vinnik case showed Greece will arrest but then spent years processing competing extradition requests from Russia and France, extradited to France, not U.S. (2022). Viable for arrest; unreliable for extradition completion.	TIER 2, ARREST ONLY. Use for disruption and detention while primary extradition proceedings run through a more reliable jurisdiction. Do not pre-position as sole extradition pathway.

A.3 Tier 3, Limited Viability

Jurisdiction	Actor Presence	Key Problem	Use
Cyprus	Very high, major Russian business and residential jurisdiction; significant Russian asset holding post-2022 sanctions evasion	Deep Russian economic integration creates political obstacles to cooperation. Low active cooperation track record despite bilateral U.S. extradition treaty.	FINANCIAL DOCUMENTATION ONLY. Most valuable as a financial registry and asset mapping jurisdiction. Do not pre-position for arrest.

Jurisdiction	Actor Presence	Key Problem	Use
Hungary	Moderate, Budapest transit point; some Russian actor presence	Orban government's Russia policy makes this jurisdiction actively unreliable for Russia-linked cases regardless of EU membership and treaty status. Has blocked EU sanctions measures.	AVOID FOR RUSSIA-LINKED CASES. Do not pre-position here.
Turkey	Very high, Istanbul and Antalya are among the most significant Russian actor transit and residence jurisdictions post-2022 sanctions; Turkish financial system actively used for sanctions evasion	Independent Russia policy and economic interests create significant unpredictability. Some cases have received cooperation; others have not. Treat every Turkey-based operation as a potential compromise risk.	LIMITED, TIER 3. Useful for financial documentation and asset mapping. Do not rely on Turkish cooperation for high-priority arrest operations without current bilateral relationship assessment.

A.4 Avoid, Explicit Entries

Jurisdiction	Why Avoid
Serbia	Russia alignment and limited extradition cooperation. Serbian authorities have demonstrated willingness to alert Russian-linked subjects to law enforcement interest, making Serbia an active operational security risk. Do not pre-position, do not share case information with Serbian authorities, do not rely on Serbian cooperation for any Russia-linked operation.
Belarus	Union State, operationally equivalent to Russia. Lukashenko government will not cooperate on any Russia-linked cybercrime case. Actors based in Belarus have the same practical protection as actors based in Russia.
Armenia / Georgia	Georgia and Armenia have both produced confirmed arrests of Russian nationals and should not be treated as equivalent to Russia-aligned jurisdictions. Neither has a U.S. extradition treaty, and formal extradition frameworks are underdeveloped in both. However, operational cooperation has occurred on a case-by-case basis and should be pursued through bilateral LE channels rather than formal treaty mechanisms. Georgia is the more viable of the two post-2022: Russia-Georgia tensions have increased Georgian willingness to act, and Georgian authorities have demonstrated operational cooperation on specific cases. Armenia has also produced confirmed arrests despite its historically closer Russian alignment. Both jurisdictions carry risks of Russian pressure on local authorities and require tight operational security and limited advance disclosure. Pre-position through bilateral relationships, not MLAT. Do not treat as primary extradition jurisdictions; treat as viable arrest and temporary detention jurisdictions with

Jurisdiction	Why Avoid
	case-by-case assessment required.
Azerbaijan	No U.S. extradition treaty. Significant Russian economic and political influence. No confirmed operational cooperation on Russia/CIS cybercrime cases. Cooperation posture is unpredictable and insufficient data exists to assess viability. Do not pre-position. Monitor for changes in bilateral posture; reassess if cooperation track record develops.

A.5 Jurisdictional Quick Reference

Jurisdiction	Extradition (US)	Russia Treaty	Track Record	Travel Relevance	Tier
Germany	Yes	None	High	High	Tier 1, Viable
Netherlands	Yes	None	Exceptional	High	Tier 1, Viable
Spain	Yes	None	Good	Very High	Tier 1, Viable
France	Yes	None	Moderate	Mod-High	Tier 2, Conditional
Poland	Yes	None	Increasing	Moderate	Tier 2, Conditional
Czech Republic	Yes	None	Proven	Moderate	Tier 2, Conditional
Italy	Yes	None	Moderate	High	Tier 2, Conditional
Greece	Yes	None	Inconsistent	High	Tier 2, Arrest only
Cyprus	Yes	Close	Low	Very High	Tier 3, Finance only
Hungary	Yes (EU)	Close	Unreliable	Moderate	AVOID
Turkey	Yes	Maintained	Unpredictable	Very High	Tier 3, Limited
Serbia	Yes	Close	Unreliable	Moderate	AVOID, OPSEC risk
Belarus	None	Union State	None	N/A	AVOID, Treat as Russia
Armenia / Georgia	Partial	Variable	Insufficient	Low-Mod	Tier 2, Arrest / Bilateral Only

Jurisdiction	Extradition (US)	Russia Treaty	Track Record	Travel Relevance	Tier
Azerbaijan	None	Close	None confirmed	Low	AVOID, Insufficient Data

A.6 Legal Pre-Positioning Checklist

For each priority jurisdiction where an actor has documented travel patterns, complete the following before any arrest window opens:

- Sealed indictment or arrest warrant filed in U.S. jurisdiction, this is the prerequisite for all pre-positioning
- MLAT request submitted to target jurisdiction, initiate 60-120 days before expected travel window
- Liaison relationship activated: FBI Legal Attaché (Legat) or DEA country office informed and briefed, do not rely on cold MLAT submissions alone
- Provisional arrest framework confirmed with host country judicial authority, know exactly what documentation the host country judge will require
- Travel pattern intelligence current, verify that the subject is still traveling to the target jurisdiction; patterns change
- Competing extradition risk assessed, is Russia likely to file a competing extradition request? If yes, prepare legal response framework in advance
- Operational security: limit knowledge of pre-positioning to minimum necessary personnel in both U.S. and host country, leaks have burned arrest windows in multiple prior cases

DOCUMENT MAINTENANCE

This playbook should be reviewed and updated quarterly. Key triggers for unscheduled updates: major takedown or law enforcement action, significant actor rebrand or ecosystem restructuring, new agency alignment evidence from Russian domestic enforcement, material change in VASP or infrastructure provider compliance posture, new jurisdictional cooperation developments in Annex A jurisdictions, or identification of new FSB officer protection relationships requiring integration into Section 9 targeting.

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